

Given:

The aim of this project is to clean and preprocess the Titanic dataset by identifying and handling missing values, converting categorical data into numerical format, standardizing numerical features, detecting and removing outliers, and preparing the dataset for machine learning model development.

DATASET: [click to download titanic dataset](#)

Data Cleanin.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
df = pd.read_csv(r"Titanic-Dataset.csv")
print(df.head())
print(df.info())
print(df.isnull().sum())
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df.drop("Cabin", axis=1, inplace=True)
encoder = LabelEncoder()
df["Sex"] = encoder.fit_transform(df["Sex"])
df["Embarked"] = encoder.fit_transform(df["Embarked"])
scaler = StandardScaler()
df[["Age", "Fare"]] = scaler.fit_transform(df[["Age", "Fare"]])
plt.figure(figsize=(8,5))
sns.boxplot(data=df[["Age", "Fare"]])
plt.title("Before Removing Outliers")
plt.show()
Q1 = df["Fare"].quantile(0.25)
Q3 = df["Fare"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
df = df[(df["Fare"] >= lower) & (df["Fare"] <= upper)]
plt.figure(figsize=(8,5))
sns.boxplot(data=df[["Fare"]])
plt.title("After Removing Outliers")
plt.show()
print(df.head())
print(df.info())
```

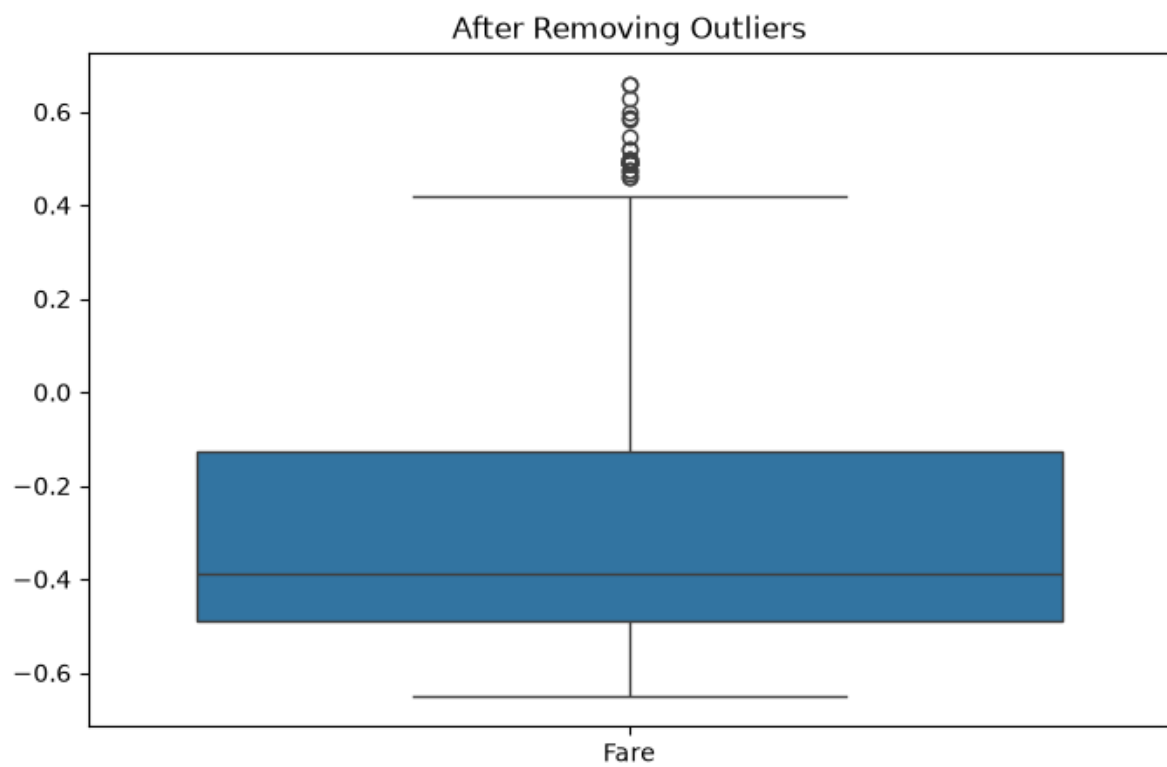
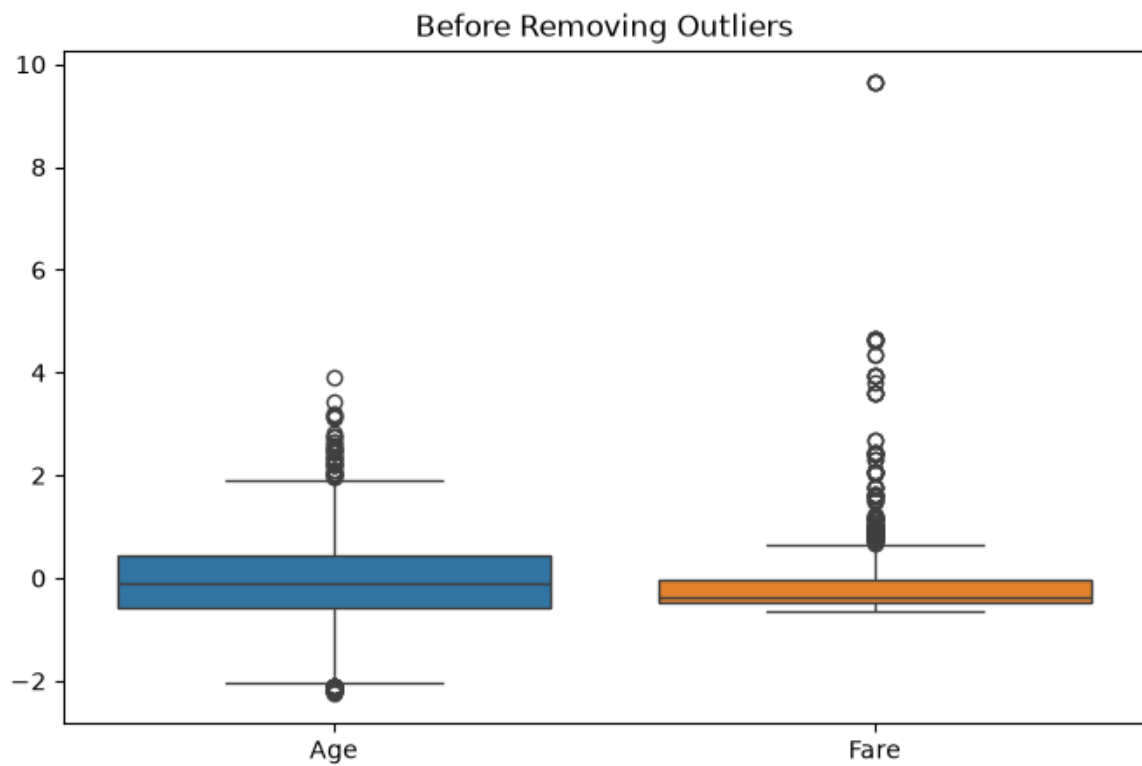
Output:

	PassengerId	Survived	Pclass	...	Fare	Cabin	Embarked
0	1	0	3	...	7.2500	NaN	S
1	2	1	1	...	71.2833	C85	C
2	3	1	3	...	7.9250	NaN	S
3	4	1	1	...	53.1000	C123	S
4	5	0	3	...	8.0500	NaN	S

```

[5 rows x 12 columns]
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null   int64
1   Survived     891 non-null   int64
2   Pclass       891 non-null   int64
3   Name         891 non-null   str
4   Sex          891 non-null   str
5   Age          714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   str
9   Fare         891 non-null   float64
10  Cabin        204 non-null   str
11  Embarked     889 non-null   str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
dtype: int64

```



	PassengerId	Survived	Pclass	...	Ticket	Fare	Embarked
0	1	0	3	...	A/5 21171	-0.502445	2
2	3	1	3	...	STON/O2. 3101282	-0.488854	2
3	4	1	1	...	113803	0.420730	2
4	5	0	3	...	373450	-0.486337	2
5	6	0	3	...	330877	-0.478116	1

```
[5 rows x 11 columns]
<class 'pandas.DataFrame'>
Index: 775 entries, 0 to 890
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  775 non-null   int64
1   Survived     775 non-null   int64
2   Pclass       775 non-null   int64
3   Name         775 non-null   str
4   Sex          775 non-null   int64
5   Age         775 non-null   float64
6   SibSp        775 non-null   int64
7   Parch        775 non-null   int64
8   Ticket       775 non-null   str
9   Fare         775 non-null   float64
10  Embarked     775 non-null   int64
dtypes: float64(2), int64(7), str(2)
memory usage: 72.7 KB
None
```